

Network Incident Root Cause Analysis: TICKET-1002

****Root Cause Analysis (RCA) Report****

****TICKET-1002: Intermittent Audio Loss between London and NY Offices****

****EXECUTIVE SUMMARY****

On [Date], a Root Cause Analysis (RCA) was conducted to investigate the intermittent audio loss between London and NY offices, as reported in TICKET-1002. The analysis revealed that the issue was caused by a misconfigured firewall rule for the RTP port range 16384-32767 on both the London and NY firewalls. This report provides a detailed overview of the incident, the technical root cause, resolution steps, and preventative measures to prevent recurrence.

****INCIDENT TIMELINE****

Time	Event
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00:16:44.454	CMProcMon - Entered Router Verification
00:16:44.454	CMProcMon - Exited Router Verification
00:16:46.454	CMProcMon - Entered Router Verification
00:16:46.454	SdlStat - Period: 6s #Lines: 23 #Bytes: 4129 Total Number of Buffers: 8000 Free LWM: 7994 Free LWM(total): 5246
00:16:52.618	SdlError - Did not receive Poll Response From NodeId: 11, AppId: 100, TCPAddr[:0]
00:16:52.618	SdlError - Did not receive Poll Response From NodeId: 11, AppId: 200, TCPAddr[:0]

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| 00:16:52.618 | SdlError - Did not receive Poll Response From NodeId: 10, AppId: 100, TCPAddr[:0]

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| 00:16:52.618 | SdlError - Did not receive Poll Response From NodeId: 10, AppId: 200, TCPAddr[:0]

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| 00:17:02.625 | SdlError - Did not receive Poll Response From NodeId: 11, AppId: 100, TCPAddr[:0]

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| 00:17:02.625 | SdlError - Did not receive Poll Response From NodeId: 11, AppId: 200, TCPAddr[:0]

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| 00:17:02.625 | SdlError - Did not receive Poll Response From NodeId: 10, AppId: 100, TCPAddr[:0]

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| 00:17:02.625 | SdlError - Did not receive Poll Response From NodeId: 10, AppId: 200, TCPAddr[:0]

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****TECHNICAL ROOT CAUSE****

The technical root cause of the intermittent audio loss between London and NY offices was a misconfigured firewall rule for the RTP port range 16384-32767 on both the London and NY firewalls. This rule was not allowing necessary traffic to pass through, causing the audio loss. The issue was similar to previous incidents (TICKET-1039, TICKET-1080, TICKET-1038), where the resolution was to fix the firewall rule for the RTP port range 16384-32767.

****RESOLUTION STEPS****

To resolve the issue, the following steps were taken:

1. Verify the firewall rules for the RTP port range 16384-32767 on both the London and NY firewalls.
2. Ensure that the RTP port range is correctly configured on the Cisco CUCM and SBC systems.
3. Update the firewall rules to allow necessary traffic to pass through.

****PREVENTATIVE MEASURES****

To prevent recurrence of this issue, the following preventative measures were implemented:

1. Regularly review and update firewall rules to ensure they are correctly configured.
2. Implement a change management process to ensure that changes to firewall rules are properly documented and tested.
3. Conduct regular security audits to identify potential vulnerabilities and address them before they become issues.

****CONCLUSION****

The Root Cause Analysis (RCA) for TICKET-1002 revealed that the intermittent audio loss between London and NY offices was caused by a misconfigured firewall rule for the RTP port range 16384-32767 on both the London and NY firewalls. The issue was resolved by updating the firewall rules to allow necessary traffic to pass through. To prevent recurrence, regular review and update of firewall rules, implementation of a change management process, and regular security audits were implemented.